EXHIBIT C

United States v. CEMEX, Inc.
Civil Action No. 09-cv-00019-MSK-MEH
U.S. District Court for the District of Colorado

Expert Report of Kevin Davis, Ph.D.

May 10, 2011

Kevin Davis

ON BEHALF OF CEMEX, INC.

Table of Contents

1. Background

This section addresses information required under relevant federal rules, describes the purpose of this report and presents a summary of conclusions.

1.1 Information Required by the Federal Rules of Civil Procedure

Appendix A contains a list of documents I considered during the preparation of this report.

Appendix B describes my qualifications.

Appendix C describes my relevant publications during the past ten years.

I have provided no previous testimony as an expert witness at trial or by deposition during the last four years.

I am being compensated at rate of $330/hr for review of reports, deposition transcripts and related exhibits, survey of related work available in the public literature and from professional contacts, and report preparation.

The opinions expressed in this report are my own and are based on the data and facts available to me at the time of writing. Should additional relevant or pertinent information become available, I reserve the right to supplement the discussion and findings in my report.

1.2 Purpose of Report

This report has been prepared at the request of CEMEX with the following objectives:

- To provide a description of the theoretical background related to NOx emissions from the coal-fired Lyons cement kiln system (rotary kiln and calciner)
- To evaluate the potential impact on NOx emissions of the implementation of oxygen enhancement in the tertiary air stream
- To evaluate the potential impact on NOx emissions of the 1997 calciner refractory modifications above the coal burners
- Given available information, to consider the impact of the spring 1997 and spring 1998 testing campaigns in evaluating the likely impact of the implementation of oxygen enhancement in the tertiary air stream

1.3 Summary of Conclusions

(1) Based upon the specific conditions present in the Lyons plant calciner and the details of the approach used to enhance the oxygen content of the tertiary air, it was reasonable to expect the system to result in lower NOx emissions at increased clinker production rates.

(2) Based upon the specific conditions present in the Lyons plant calciner and the details of the approach used to modify the refractory walls immediately above the calciner coal burners, it was reasonable to expect the system to result in lower NOx emissions at increased clinker production rates.

(3) Based upon available documentation, the 1997 test campaign involving multiple levels of tertiary air oxygen enhancement appears to support the expectation of reduced NOx emissions at increased clinker production rates for all tested levels of oxygen injection rate.

(4) Based upon available documentation, the 1998 test campaign with and without tertiary air oxygen enhancement and five kiln exit oxygen concentrations provided the plant with additional experience regarding the impact of calciner improvements and suggests that, under the tested operating conditions, the impact of tertiary air oxygen enhancement on NOx emissions was insignificant under relevant kiln exit oxygen concentrations.

(5) Based upon (1) the complexity and sensitivity of NOx emissions in a coal-fired cement kiln, (2)the magnitude and nature of the fluctuations in the historical NOx data collected at the Lyons plant, (3) the unique combustion effects of the calciner improvements, a simple application of emissions factors would be of little value in evaluating potential NOx impacts related to the tertiary air oxygen enhancement system and calciner geometry improvements considered herein. Consideration of NOx emissions, unlike other pollutant emissions such as SOx, mercury, or particulate matter (PM), cannot be made on the basis of fuel properties and firing rates, much less on simple emissions factors related to the amount of clinker produced.

(6) Based upon the combination of the March 1997 testing, the day-to-day experience of operating with tertiary air OE for several months, and the April 1998 testing, it would be reasonable to assume that the plant had an acceptable basis for believing that the addition of the tertiary air OE system and the calciner refractory project would not increase NOx emissions and, under typical conditions, would decrease NOx emissions, even with increased production. The actual emissions data for the 24 month period before and after the calciner improvements support the plant's conclusion and indicate

that the changes were associated with a decrease in absolute NOx emissions while increasing clinker production.

2 NOx Chemistry during Coal Combustion

NOx chemistry in a coal-fired precalciner kiln is strongly impacted by the details of the combustion process and is the result of multiple, interacting, and often non-linear mechanisms. Consideration of NOx emissions, unlike other pollutant emissions such as SOx, mercury, or particulate matter (PM), cannot be made on the basis of fuel properties and firing rates, much less on simple emissions factors related to the amount of clinker produced. As discussed in Section 4, emissions factors are highly variable and it would be unwise to make conclusions related to NOx emissions based on the assumption that these factors would remain constant when changes involving NOx chemistry are under consideration.

2.1 Mechanisms of NOx formation during coal combustion

The generation of calcium silicates (clinker) used for the subsequent production of portland cement requires a substantial amount of energy to be delivered to a mixture of raw materials (rawmix) including calcium carbonate (in the form of limestone) and alumino-silicates (typically in the form of clay/shale). The use of a rotary kiln effectively creates the high temperature, well mixed conditions in the solids bed that are necessary for production of a high quality clinker. However, in achieving the desired temperature (~2650 °F) needed for the sintering transformation, even hotter (~3350-3650 °F) fossil-fuel-fired flames are required in the rotary kiln (although not in the calciner).[1] In addition, acceptable clinker quality can only be realized when the transformations occur under conditions with excess oxygen. The combination of high flame temperatures and oxygen-rich gases have the potential to result in high concentrations of nitrogen oxides (NOx) in kiln exhaust gases.

Theoretically there are a number of mechanisms[2] resulting in NOx formation in coal combustion systems, however, in precalciner cement kilns, the most important are thermal fixation of atmospheric nitrogen (thermal NOx) and reactions involving the nitrogen inherently present in the fuel itself (fuel NOx).

2.1.1 Thermal NOx

The predominant oxides of nitrogen (NOx) are NO and NO2. Both can be generated during combustion; however, NO is the more important of the two species in the energy conversion

[1] K. Peray, The Rotary Cement Kiln, CHS Press, 1986, p. 141.

[2] Other mechanisms such as prompt NOx (relevant for applications in which NOx concentrations are very low) and feed NOx (relatively limited in the high heating rate environment of a calciner) are considered to be minor for the purposes of this discussion.

reactions that occur in a cement kiln. The well-known Zeldovich mechanism[3] provides a basis for describing the amount of thermal NO formed:

$$N_2 + O \leftrightarrow NO + N$$

$$O_2 + N \leftrightarrow NO + O$$

At high enough temperatures the typically stable oxygen molecules in air dissociate to highly reactive oxygen atoms (radicals), which then attack typically stable nitrogen molecules forming NO and nitrogen radicals. Similarly, nitrogen radicals attack the oxygen atoms, again creating NO, as well as additional oxygen radicals.

As illustrated in Figure1, thermal NOx formation depends strongly upon gas temperature and to a lesser extent upon oxygen concentration.[4]

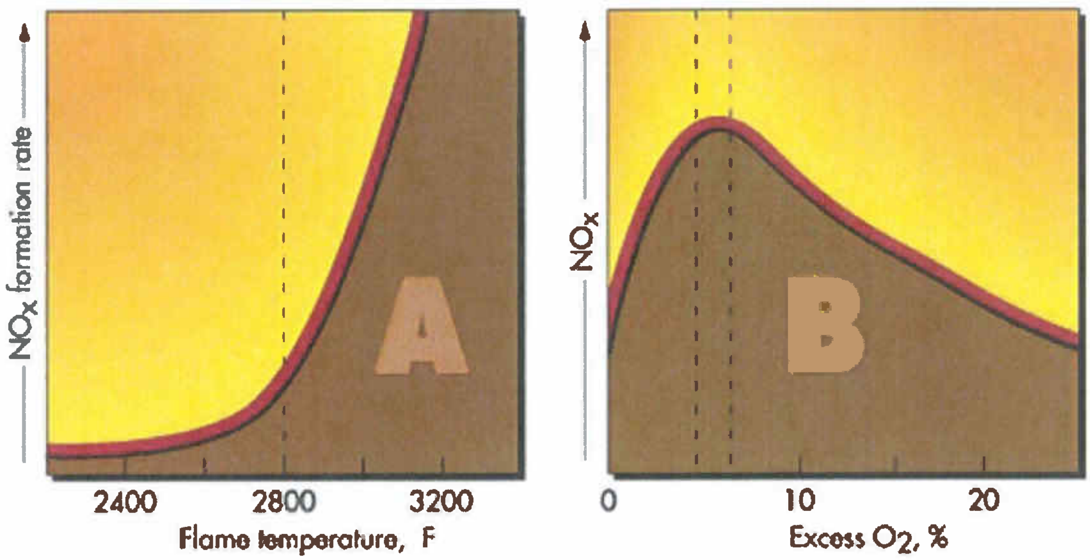

Figure 1. Dependence of NOx formation on temperature and excess oxygen concentration.[5,6]

As a quantitative example, in gas flames where fuel NO does not complicate interpretation of results, a change of 190 °F has been observed to impact the NO by a factor of two.[7] In the range of excess O_2 typically targeted for combustion equipment (below 6%), flame temperature

[3] Y. Zeldovich "The Oxidation of Nitrogen in Combustion Explosions," Acta Physicochima, 21, 577 (1946).
[4] I. Glassman, Combustion, Academic Press, 2nd ed., 1987, p. 330.
[5] http://www.alentecinc.com/papers/NOx/The%20formation%20of%20NOx_files/The%20formation%20of%20NOx.htm
[6] Although NOx formation rate is more directly related to near-flame oxygen concentration, excess oxygen (as shown in the second plot in Figure 1) is typically related to near-flame oxygen concentration and is more easily measured/understood.
[7] http://www.ftek.com/en-US/products/apc/low-nox-burners/

is a relatively weak function of excess O_2, and NOx decreases significantly as excess O_2 is reduced below 4%.

In the main flame of a coal-fired cement kiln, temperatures are well over 3000 °F and thermal NOx dominates overall NOx formation.[8] The highly non-linear nature of thermal NOx formation should be noted, as this behavior is an important factor in understanding why cement kiln NOx emissions can vary dramatically with relatively small changes in operation.

Numerous approaches have been developed and successfully applied to keep peak flame temperatures down and to mitigate the formation of thermal NOx in flames including staged combustion, water injection, flue gas recycle, and combustion air preheat reduction (e.g., in the clinker cooler).

2.1.2 Fuel NOx

Fossil fuels such as coal, oil and natural gas contain varying amounts of nitrogen. Coal is relatively high in nitrogen, containing roughly 1.5% nitrogen (by weight), while natural gas contains practically no nitrogen other than the diatomic form that is the major constituent of air. The reactions involving fuel nitrogen are highly complex and their theoretical understanding, although extensively studied, remain under development. Fuel nitrogen can evolve due to a combination of decomposition, oxidation, gasification and other reactions. However, in general, coal-bound nitrogen will produce NOx as a result of both pyrolysis and char oxidation and the fractional conversion to NOx of the nitrogen released during each phase varies widely depending upon a number of factors including coal composition, local gas NOx concentration, two-phase flow and mixing, and heat transfer.[9,10,11] The forms of nitrogen in coal are complex and varied. Figure 2 illustrates details of the nitrogen-containing functionalities and estimates of their relative composition in coal.

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[8] STAPPA/ALAPCO "Controlling Nitrogen Oxides Under the Clean Air Act: A Menu of Options", Cement Kilns Chapter, July 1994.

[9] D. Pershing, J. Wendt, "Relative Contributions of Volatile Nitrogen and Char Nitrogen to NOx Emissions from Pulverized Coal Flames," Ind. Eng. Chem. Process Des. Dev., 18, No. 1 (1979).

[10] K. Davis and C. Senior "Fuel effects on NOx control," NOx Round Table & Expo, Charlotte, NC, January, 2006.

[11] K. Davis "Low NOx Burners Tutorial: Fuel Effects" 31st International Technical Conference on Coal Utilization & Fuel Systems, Clearwater, FL, May 2006.

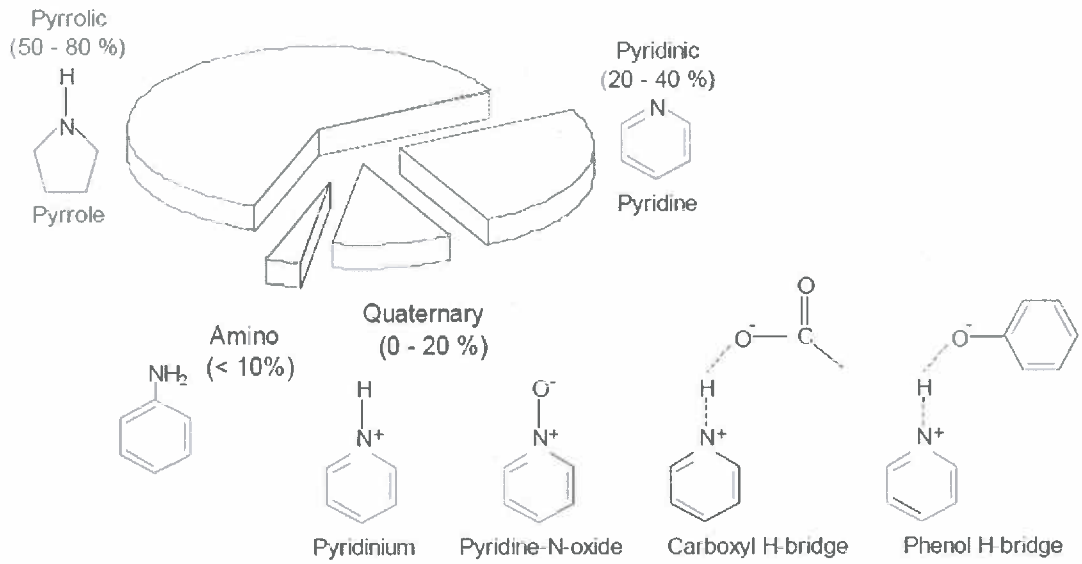

Figure 2. Distribution of nitrogen-containing functionalities in coal.[12]

This complexity in understanding fuel NOx is further compounded by the interactions between the simultaneous interacting reaction pathways and their multi-step nature as illustrated in Figure 3. However, in a cement kiln, the eventual NOx produced through the fuel mechanism is limited by the amount of nitrogen in the fuel and the mitigating effect of the typically high concentrations of NOx generated thermally. The relatively minor role of fuel NOx in cement kilns is also supported by the observation that NOx emissions from cement kilns are lower when fired with coal as opposed to natural gas.[13] Natural gas flames are hotter than coal flames even though the adiabatic flame temperature is higher for coal than for natural gas.[14]

[12] A. Molina, Evolution of Nitrogen During Char Oxidation. Ph.D. Dissertation, The University of Utah, 2002, p. 3.
[13] STAPPA/ALAPCO "Controlling Nitrogen Oxides Under the Clean Air Act: A Menu of Options", Cement Kilns Chapter, July 1994.
[14] B. Neuffer, M. Laney "Alternative Control Techniques Document Update – NOx Emissions from New Cement Kilns," EPA Report 453/R-07-006, November 2007.

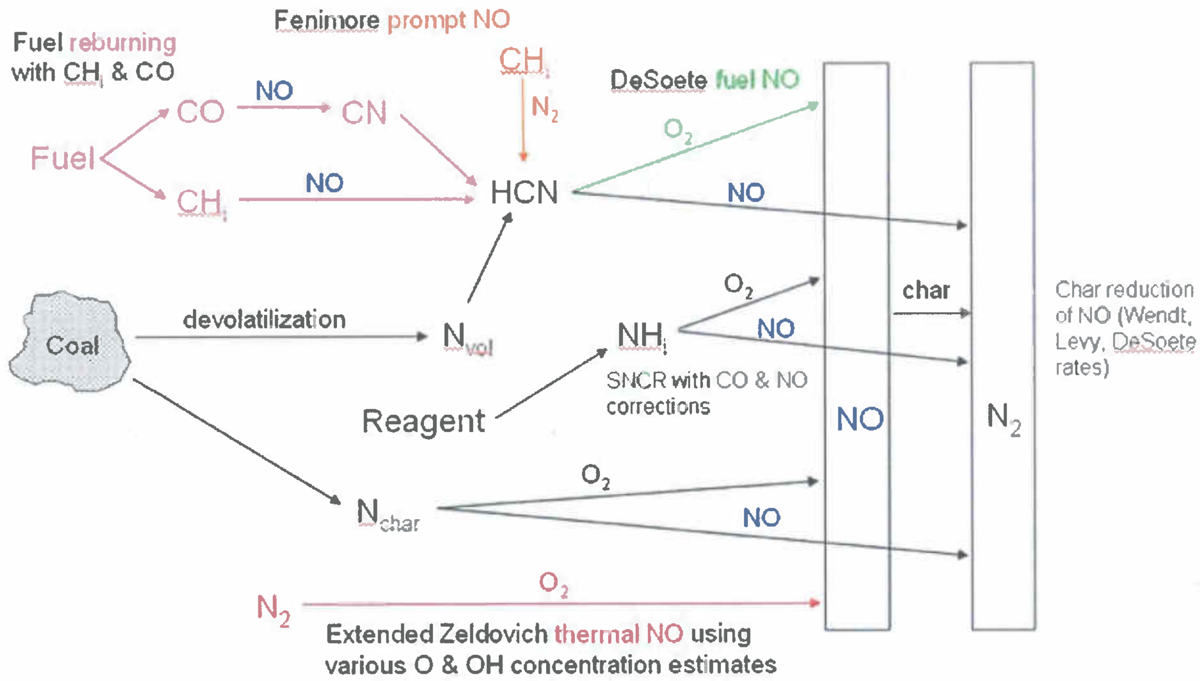

Figure 3. Mechanistic description of the chemical pathways affecting NOx evolution during coal combustion.[15]

2.2 Approaches/mechanisms for NOx destruction

Developing a basis for understanding NOx chemistry in the calciner (as opposed to the rotary kiln itself) can be aided by consideration of NOx control approaches that have been developed for post-rotary kiln application. This section will provide that background, which in turn will inform the discussion of the calciner improvement projects in Section 3 below.

The number of commercially available technologies for post-combustion NOx control for cement kilns is relatively small in comparison to the range of choices for other combustion equipment such as boilers and incinerators. However, improvements in system efficiency have been substantial in recent decades and there has been a corresponding improvement in NOx emissions per ton of clinker. The increased use of PH/PC kilns has not only improved efficiency, but also results in a reduction in the amount of fuel fired in the kiln burner. This reduction results in a correspondingly lower flame temperature and less NOx formation. There are also NOx control technologies such as Selective Non-Catalytic Reduction (SNCR) and Staged Combustion in the Calciner (SCC) that are experiencing increasing use, as described below. This section will also discuss the potential of two NOx control approaches, Fuel-lean reburn and oxygen injection, that are relevant when considering the potential impacts of (1) adding

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[15] K. Davis and C. Senior "Fuel effects on NOx control," NOx Round Table & Expo, Charlotte, NC, January, 2006.

calciner oxygen injection and (2) incorporating calciner geometry modifications to enhance mixing.

2.2.1 Selective Non-catalytic Reduction

As of July 2000, commercial experience with SNCR in cement kilns was limited.[16] However, the largest US based supplier of SNCR systems, Fuel Tech, had conducted a number of demonstrations and by 2007 there were several commercial installations in the US and several more in the planning stage according to the US EPA.[17] According to STAPPA ALAPCO (now National Association of Clean Air Agencies), NOx removal efficiencies in precalciner kilns are reported at 30-70 percent.

The experience with SNCR performance in cement kilns illustrates the potential for NOx reduction as exhaust gases pass through the calciner of a PH/PC kiln. Due to geometry and process considerations, the calciner can provide post-reagent injection residence time at an appropriate temperature and composition for SNCR-related NOx chemistry; most of the rotary kiln is too hot or too stratified and as the NOx-containing exhaust gases pass downstream of the calciner, the preheating process will create temperatures that are too low. SNCR providers emphasize the importance of addressing limitations in (1) residence time at sufficiently high temperature[18] and (2) injection of reducing reagent into a well mixed gas stream.[19] As will be discussed subsequently, these two issues can also play an important role in understanding the impacts of NOx reduction involving "reburning" chemistry.

2.2.2 Staged combustion in the calciner (SCC)

The concept of fuel (and/or air) staging for NOx control can be considered at many levels. In the near-burner region, the mixing of fuel and air can be carefully controlled through proper low NOx burner design such that both fuel and thermal NOx formation mechanisms can be slowed. In addition, the flame can be tailored so that the near burner NOx that is formed can be subsequently reduced in locally high temperature, rich pockets.[20] In the post-combustion regime, fuel staging can also be applied such that a significant fraction of the total fuel is added downstream of the main flame(s) at lower temperatures. Across a broad range of furnace and

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[16] L. Lin, M. Knenlein, "Cement Kiln NOx Reduction Experience Using the NOxOUT® Process," Proceedings of 2000 International Joint Power Generation Conference, Miami Beach, Florida, July 23-26, 2000.
[17] B. Neuffer, M. Laney "Alternative Control Techniques Document Update – NOx Emissions from New Cement Kilns," EPA Report 453/R-07-006, November 2007.
[18] B. Neuffer, M. Laney "Alternative Control Techniques Document Update – NOx Emissions from New Cement Kilns," EPA Report 453/R-07-006, November 2007.
[19] E. Hansen, F. Lockwood "Enhancing SNCR Performance," International Cement Review, December 2006.
[20] K. Davis, M. Bockelie, P. Smith, M. Heap, R. Hurt, J. Klewicki "Optimized Fuel Injector Design for Maximum In-furnace NOx Reduction and Minimum Unburned Carbon," First Joint Power and Fuel Systems Contractors Conference, Pittsburgh Energy Technology Center, July 1996.

boiler applications this is often referred to as reburning and has many similarities to the staged combustion approach commonly used in the calciner of precalciner kilns.[21]

2.2.2.1 Reburning for NOx control

Reburning is a well know technique for reducing NOx emissions related to coal combustion and is based on laboratory work performed in the early 1970's by Wendt and coworkers of Shell Development.[22, 23] Most implementations of this technology have used natural gas as the reburning fuel; however, coal reburning systems are also well developed and have been installed commercially.[24] Reburning involves the injection of a hydrocarbon to create a fuel-rich region in which reactive intermediates can react with NO converting it to species such as HCN and NH_i. These nitrogen-containing species can then be converted to N_2 given appropriate conditions of stoichiometry, temperature, and residence time. The locally fuel-rich regions where the conversion to N_2 has occurred are then subsequently oxidized under conditions less likely to create NOx.

A detailed understanding of coal reburning is still evolving; however, it is clear that coal reburning involves gas-phase reburning by pyrolysis products,[25] heterogeneous reactions with char[26] and possibly a third mechanism involving gas-phase char oxidation products.[27] The effectiveness of coal reburning in NOx reduction is strongly impacted by the stoichiometry, temperature, and residence time of the reburning zone as well as mixing and stoichiometry of the reburn fuel jet.[28] In addition, numerous studies involving approaches to enhance near-coal-injector volatile yield have also illustrated the improved performance of reburn with reburn coal volatile yield. These approaches have included the following:

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[21] B. Neuffer, M. Laney "Alternative Control Techniques Document Update – NOx Emissions from New Cement Kilns," EPA Report 453/R-07-006, November 2007.

[22] A. Kokkinos, J. Cichanowicz, D. Eskinazi, J. Stallings, G. Offen, "NOx Controls for Utility Boilers: Highlights of the EPRI July 1992 Workshop," J. Air and Waste Management Association, 42, pp. 1498 – 1505, 1992.

[23] J. Wendt, C. Sternling, M. Matovich. "Reduction of Sulfur Trioxide and Nitrogen Oxides by Secondary Fuel Injection," 14th Symposium (International) on Combustion, pp. 897-904. The Combustion Institute, 1973.

[24] US Department of Energy "Reburning Technologies for the Control of Nitrogen Oxides Emissions from Coal-Fired Boilers," Topical Report Number 14, May 1999.

[25] J. Mereb, J. Wendt, Air Staging and Reburning Mechanisms for NOx Abatement in a laboratory Coal Combustor, Fuel, 73, pp. 1020 – 1026, 1994.

[26] W. Chen, L. Ma, Effect of Heterogenous Mechanisms during Reburning of Nitrogen Oxide. AIChE Journal, 42, pp. 1968 – 1976, 1996.

[27] E. Eddings, K. Kelly, D. Overacker, C. Thurston "Enhanced Coal Reburning in Oxidizing Environments," DOE Final Technical Report, Project No. DE-FG26-02NT41550, 2004.

[28] S. Chen, J. McCarthy, W. Clark, M. Heap, W. Seeker, D. Pershing "Bench and Pilot Scale Process Evaluation of Reburning for In-furnace NOx Reduction," 21st Symposium (International) on Combustion, pp. 1159-1169. The Combustion Institute, 1986.

- High volatile yield/fast pyrolysis rate coal (Powder River Basin Subbituminous coal)[29]
- Oxygen enhanced reburn[30]
- Micronized coal reburn[31]

The importance of creating conditions that result in stable reburn flames with high volatile yields is further emphasized by Maringo and coworkers[32] as well as Yang and coworkers.[33] The NOx emissions benefits of a well attached reaction zone is not limited to reburn fuel injection and is more clearly explained in terms of a traditional coal burner application. Figure 4 illustrates a well attached and unattached coal flames. Low NOx burners rely upon slowing the mixing of fuel and air and creating fuel-rich regions in the volatile flame. This can result in a balancing act between flame stability and controlled mixing. If the flame is not well attached and stable, then there is significant mixing of the combustion air surrounding the coal stream with the coal stream itself prior to the release of volatile material. This creates air-rich conditions that result in greater conversion of the coal nitrogen to NOx.

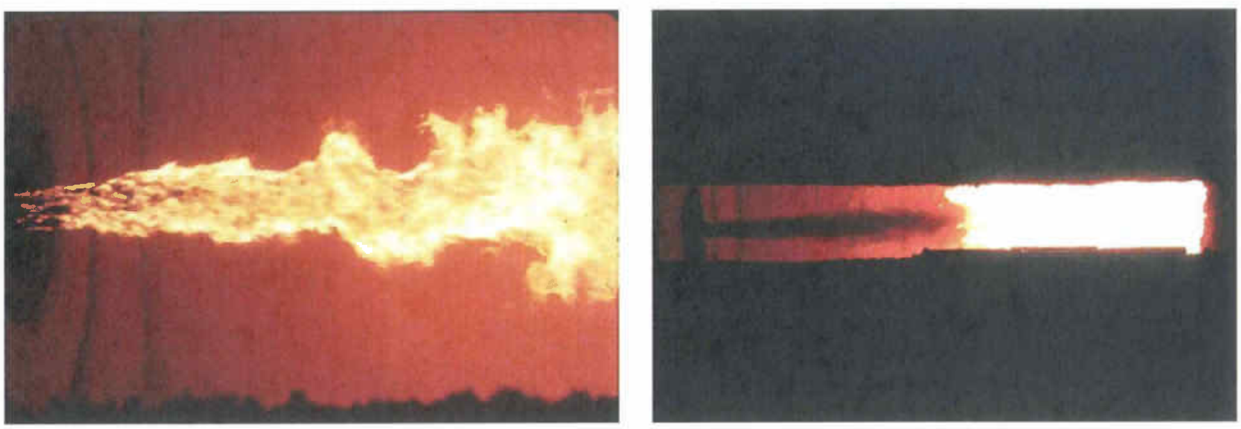


Figure 4. A well attached and unattached coal flame in a pilot-scale combustion test facility.[34]

High ash coal flames provide a clear example of the potential impact of flame attachment on NOx emissions. Davis and coworkers[35] compared the NOx emissions from a high ash feedstock

29 G. Maringo, A. Yagiela, R. Newell "Compliance Benefits of Powder River Basin Coal and Coal Reburning," preprint and abstract, ACS Fuels Division, San Diego, 1994.

30 L. Bool, S. Kobayashi "Oxygen-Enhanced Coal-Based Reburning," 2004 Conference on Reburning for NOx Control, May, 2004.

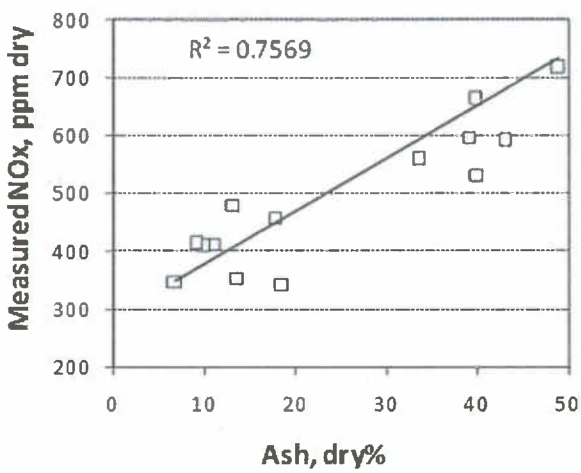
31 M. Hu, H. Tian, Z. Liu, X. Wang "Numerical Simulation of Volatiles Release for Micronized Coal Reburning," Am. Chem. Soc., Div. Fuel Chem., 49(2), p. 831, 2004.

32 G. Maringo, A. Yagiela, R. Newell "Compliance Benefits of Powder River Basin Coal and Coal Reburning," preprint and abstract, ACS Fuels Division, San Diego, 1994.

33 Y. Yang, E. Hampartsoumian, B. Gibbs "The Effects of Temperature, Mixing and Volatile Release on NO Reduction Mechanisms by Coal Reburning," 27th Symposium (International) on Combustion, pp. 3009-3017. The Combustion Institute, 1998.

34 K. Davis "Application of NOx Control Technologies," Applied Combustion Technology Short Course, Salt Lake City, UT, March, 2004.

with that of its refined product following a coal cleaning process using physical ash separation. The ash content of a solid fuel can be high enough to act as a significant heat sink thereby making it difficult to heat up the coal quickly enough to produce a stable, well attached flame near the burner. This effect can be highly non-linear as the particle heat-up time increases to a point where the volatile material is not released near enough to the burner to provide heat feedback via the recirculating combustion products to the root of the flame. Figure 5 Illustrates the noticeable increase in NOx emissions that can result from the loss of flame attachment at high coal ash content.



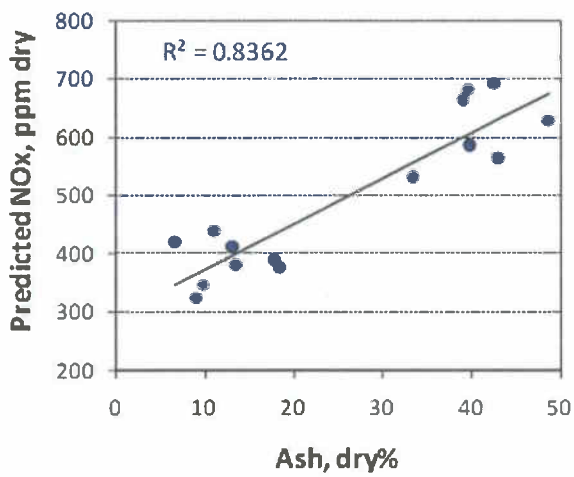


Figure 5. NOx emissions during pilot-scale combustion of high ash feedstock coals and moderate ash refined product coals – as measured (left) and as predicted by modeling (right).[36]

2.2.2.2 Fuel-lean reburn

The traditional approach to reburning for NOx control requires the creation of fuel-rich conditions downstream of the secondary/reburn fuel injection point as well as subsequent injection of burnout air to deal with products of incomplete complete combustion. A variation of the concept referred to as fuel-lean reburn is more relevant for the purpose of the discussion of conditions in an existing calciner. Le and coworkers[37] have recently suggested fuel lean reburn as an effective alternative to SNCR in cement kilns. Fuel lean reburning relies upon the creation of local reducing conditions by injection of fuel into overall fuel-lean exhaust gases containing NOx at a temperature of approximately 1650 °F. However, as explained by Miller and coworkers,[38] NOx reduction effectiveness is only weakly dependent on temperature. No

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[35] K. Davis, S. Lokare, H. Shim, S. Harding, R. Minkara "A Physical Coal Upgrading Process and Its Impact on NOx Emissions," 31st International Technical Conference on Coal Utilization & Fuel Systems, Clearwater, FL, June 2010.
[36] Ibid.
[37] E. Le, T. Reese, E. Hansen "Fuel Lean Gas Reburn in Mineral Processing Kilns," Global Cement Magazine, May 2010.
[38] A. Miller, D. Touati, J. Becker, J. Wendt "NOx Abatement by Fuel-lean Reburning: Laboratory Combustor

subsequent air is required since the reburn fuel is added to combustion products that have enough oxygen to not only consume the added fuel, but also to provide for the desired excess oxygen level at the outlet. Le and coworkers[39] point out that the primary limitation on fuel lean reburning in precalciner kilns is the lack of adequate mixing prior to the injection of reburn fuel.

3 Potential Impact of Calciner Improvements at the Lyons Plant

The concepts discussed in the previous section, particularly those involving flame attachment and fuel lean reburning, should be carefully considered with regard to potential NOx impacts of modifications affecting the calciner coal injection and reaction. During 1997 two improvements were made to the calciner at the Lyons plant that had the potential to reduce NOx emissions – (1) the installation of a system to enhance the oxygen content of the tertiary air stream entering the calciner and (2) modification of the post coal injection "combustion" chamber geometry to enhance mixing. These modifications have increased the amount of coal that can be fired in the calciner and improved the fuel efficiency of the calciner; therefore the amount of coal that must be fired in the kiln main burner to produce a given amount of clinker is reduced.[40,41] This would be expected to reduce the flame temperature in the hottest location in the process and consequently reduce thermal NOx formation in the rotary kiln. In addition, there are specific reasons to expect that both modifications could result in enhancement of calciner NOx reduction mechanisms and therefore additional decrease in overall process NOx emissions.

3.1 Calciner oxygen enhancement (OE)

Oxygen combustion in industrial applications in the U.S. has become relatively commonplace. For example, a significant fraction of commercial glass melting furnaces have been converted to oxy-fuel firing resulting in significant energy savings and production quantity/quality improvements while simultaneously reducing emissions of NOx, CO and particulates.[42] Indiscriminate use of oxygen in combustion has the potential to greatly increase flame temperatures and therefore dramatically increase NOx emissions in situations in which thermal NOx is important. However, this simplistic view of oxygen combustion does not hold true in

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and Pilot-scale Package Boiler Results," 27th Symposium (International) on Combustion, pp. 3009-3017. The Combustion Institute, 1998.

[39] E. Le, T. Reese, E. Hansen "Fuel Lean Gas Reburn in Mineral Processing Kilns," Global Cement Magazine, May 2010.

[40] H. Tseng, P. Alsop "Oxygen Enrichment of Cement Kiln System Combustion," U.S. patent 6,488,765, December 3, 2002.

[41] H. Tseng, P. Alsop "Apparatus for Oxygen Enrichment of Cement Kiln System," U.S. patent 6,688,883, February 10, 2004.

[42] H. Kobayashi, R. Prasad "A Review of Oxygen Combustion and Oxygen Production Systems," Proceedings of Forum on High Performance Industrial Furnace and Boiler, Tokyo, Japan, March 8-9, 1999.

many practical applications and the potential impact of oxygen injection on NOx emissions must be performed on a case-case basis.

Strategic use of oxygen has been successfully implemented or shown promise in a number of coal combustion applications including oxygen enhanced reburn[43] and tertiary stream oxygen enrichment.[44] These results indicate that oxygen enhancement can improve the combustion and NOx emissions of staged fuel injection and that the oxidation of fuel-rich products as they mix with tertiary air can be accomplished without a negative impact on NOx emissions. As discussed in Section 2.2.2.1, tertiary air OE can improve flame stability/attachment resulting in more rapid heating of the coal particle, which not only enhances the volatile yield available for reburning but can also reduce the NO formation reaction by releasing the coal-bound nitrogen in elemental form, rather than as NO.

NOx data related specifically to the impact of oxygen injection into the calciner of a cement kiln is not publicly available. However, in the spring of 1997, equipment for testing OE was acquired by the Lyons plant. During a test campaign, various implementations were investigated and, following the testing, injection of oxygen into the tertiary air was identified as the most desirable approach. The initial implementation used following the 1997 testing involved the injection of oxygen directly into the tertiary air ductwork near the kiln discharge hood. Subsequently, oxygen was injected into the swirl chamber immediately downstream of where the tertiary air duct enters. The point of injection remained upstream of the swirl chamber. Therefore it is reasonable to expect that this change would not have resulted in any impact on the effectiveness of the OE process on increasing production or lowering overall NOx emissions.

Prior to the use of OE, the coal burners[45] in the calciner were supplying the heat for the highly energy demanding calcination reaction occurring in the rawmix and, as indicated by the problem with continued hydrocarbon/CO reactions in the preheater, were not resulting in complete combustion.[46] The addition of oxygen into the tertiary air stream would be expected to enhance the stability/attachment of the coal flames and, at these relatively mild temperatures, where thermal NOx formation rates would be very low, it would be reasonable to assume that, for a given coal firing rate, both overall NOx and hydrocarbon/CO emissions could be reduced while clinker production increased.

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[43] L. Bool, S. Kobayashi "Oxygen-Enhanced Coal-Based Reburning," 2004 Conference on Reburning for NOx Control, May, 2004.

[44] O. Marin et al "Low-oxygen Enrichment in Coal-fired Utility Boilers," 28th International Technical Conference on Coal Utilization & Fuel Systems, Clearwater, FL, May 2003.

[45] The terminology "burner" in this instance could be confusing as most burners also have closely associated "combustion air." The term fuel injector would be a reasonable term as well.

[46] Fuel and Combustion Technology International "Southwestern Portland Cement Company Physical Model Study of Precalciner," Report RM 546, August 2, 1996, CMXLYO00453793.

Due to the highly non-linear impact of operating conditions and coal properties on NOx emissions, quantitative verification of the impact of equipment modifications on NOx emissions from a cement kiln can be very challenging. Based upon the deposition of the Plant Manager at the time, it appears that a reasonable effort was made to include CEM NOx emissions data in the OE test campaign performed in March 1997.[47] This exhibit suggests that NOx emissions data were collected for a "Base" condition without oxygen injection and with four flow rates of oxygen. All five conditions were evaluated for periods of at least 21 hours. (Under the "Base" condition data was collected for 266 hrs.) A plot of NOx (lb/hr) versus nominal oxygen mass flow rate (tons/day) is shown in Figure 7.[48]

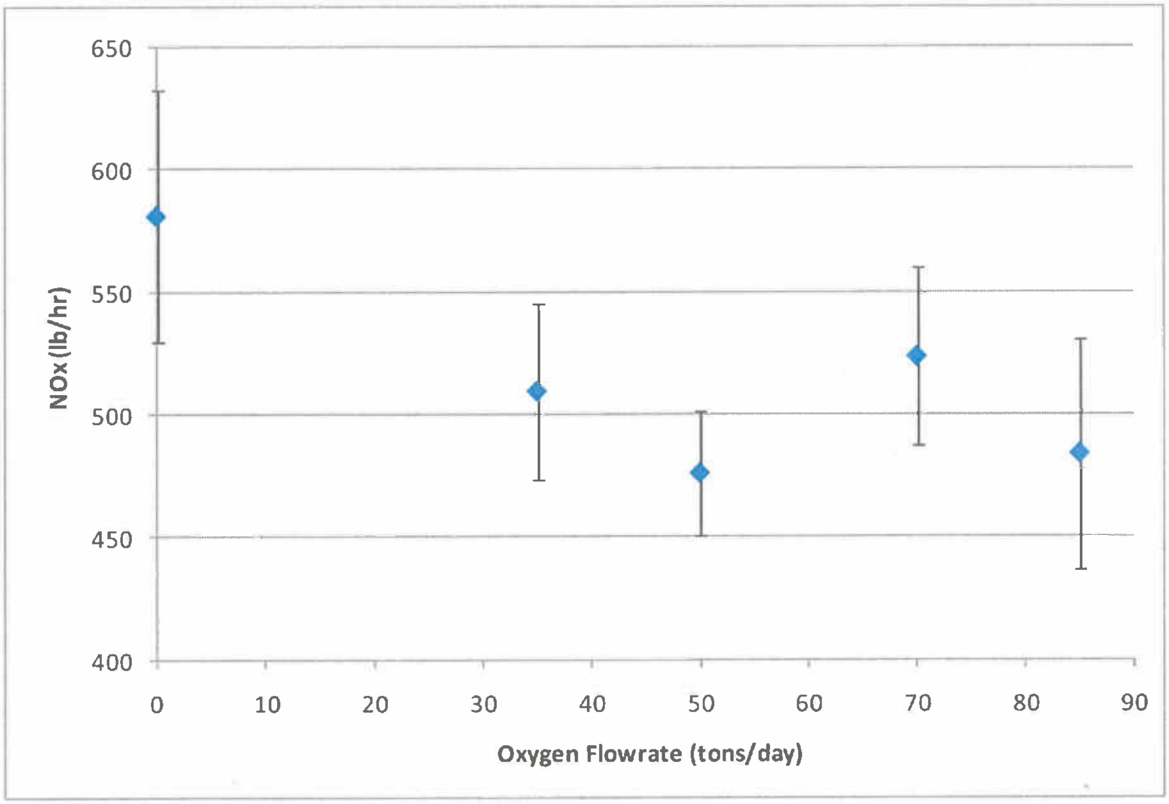

Figure 7. NOx emissions during the March 1997 testing with oxygen injection. (The error bars shown in the plot represent +/- one standard deviation.[49])

The standard deviations determined from the NOx emissions data appear quite good in light of the sensitivity issue mentioned previously. As indicated by this plot, the bulk of the data with oxygen injection is below the bulk of the data during operation without oxygen injection. This type of dependence – high baseline NOx emissions, a significant drop in NOx emissions at 35

[47] Deposition of John Lohr, October 11, 2010, Exhibit 208, CMXLYO00612399.
[48] Test data in an excel spreadsheet describing this data was also available (CMXLYO00612454). Trends and conclusions of significance are similar regardless of the source.
[49] Deposition of John Lohr, October 11, 2010, Exhibit 208, CMXLYO00612399.

tons/day and relatively modest changes beyond that, is consistent with the previous discussion involving the step change associated with flame attachment. In summary, this testing supports the potential for improvement in NOx emissions with the use of oxygen injection.

3.2 Calciner geometry improvement for mixing enhancement

Shortly after the testing of oxygen injection performed in the spring of 1997, a modification to the lower conical section of the calciner combustion chamber was completed. This modification was a relatively simple change to the existing conical section above the burner such that an eight foot high constant diameter conical section is formed.[50] This modification appears to be scaled directly from one of two modifications suggested by Fuel and Combustion Technology International[51] (FCT) to improve fuel/air mixing by improving the recirculation flow patterns within the combustion chamber. FCT's recommendations were based on a series of cold flow modeling tests and the specific geometry installed at Lyons was the least obtrusive of the two recommended approaches.

As indicated by the FCT report, calciner burners must operate under challenging conditions and, under extreme circumstances, can "flame out"(become seriously detached as shown in Figure 4). The modifications made to the refractory immediately above the burners should have resulted in improved recirculation of hot coal combustion products to the root of the flame for improved flame attachment and stability. As with OE, the improved flame stability should result in less intimate mixing of the oxygen-containing gases into the base of the fuel-rich NOx reducing region. Again, it is reasonable to expect this modification to result in improved fuel efficiency and clinker production with lower emissions of hydrocarbons, CO and NOx.

3.3 April 1998 testing

Although the FCT study involving improvements to the calciner geometry was completed prior to the March 1997 testing, the modifications were not completed until later in 1997. Therefore, during the subsequent testing in April of 1998, both OE and combustor geometry changes were in place.[52] The available documentation related to this test campaign is limited. However, it appears that a series of two hour tests involving one oxygen flow rate and five excess oxygen

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[50] Deposition of Stephen Goodrich, Exhibit 363, December 9, 2010.

[51] Fuel and Combustion Technology International "Southwestern Portland Cement Company Physical Model Study of Precalciner," Report RM 546, August 2, 1996, CMXLYO00453793.

[52] Both calciner improvements (OE and combustor geometry) result in enhanced flame attachment, which has the potential to reduce both "burnover" and NOx emissions; OE, by enhancing the reactivity of the oxidant for entrainment into the root of the flame and geometry improvement, by creating improved recirculation patterns for entrainment of hot combustion products to the root of the flame. While some of the potential NOx-related benefit of OE will overlap with that of the geometry modifications, it is reasonable to conclude that there will be an independent incremental benefit associated with both improvements and therefore the qualitative indications of the 1997 testing remain valuable.

concentration levels were completed. As discussed in Section 2.1.1, the concentration of oxygen available for the high temperature kiln flame (as reflected in the kiln exit O_2) is a factor in NOx emissions and as this variable was not explored during the 1997 testing, this effort was a reasonable supplement to the earlier work.

Based on the limited information, it is difficult to draw detailed conclusions related to NOx emissions. However, the data presented by Boardman[53] does appear to indicate that (1) at typical kiln exit oxygen concentrations (≥2.5% as indicated by the expert report of Hofmann[54]) there was no effect of tertiary air OE and (2) at low rotary kiln exit oxygen concentrations (≤ 2.0%) tertiary air OE results in an increase in absolute NOx emissions. It should also be noted that the comment in the U.S. expert report of Walter Greer (at page 45) that "an increase in oxygen concentration at the feed end of the rotary kiln from 1% to 3% should tend to cool the flame and concurrently reduce NOx generation" is likely based on simplistic assumptions and that in practice the flame temperatures referenced (4470F at 1% and 4018F at 3%) are not realistic temperatures that should be used for NOx emissions predictions. In practical coal combustion applications, NOx emissions will typically increase significantly when excess oxygen is increased from 1-3%. A detailed discussion of this issue is presented by Peray.[55]

3.4 Summary

Based on the combination of the March 1997 testing, the day-to-day experience of operating with tertiary air OE for several months, and the April 1998 testing, it would be reasonable to assume that the plant had an acceptable basis for believing that the addition of the tertiary air OE system and the calciner refractory modification were not increasing NOx emissions and under typical conditions might decrease NOx emissions. The actual emissions data for the 24 month period before and after the calciner improvements, as shown in Table 5.1 of Hofmann's report,[56] support the plant's conclusion and indicate that the changes were associated with a decrease in absolute NOx emissions while increasing clinker production.

4 Historical NOx Emissions Data and Emissions Factors

As discussed in this report, for the conditions in the calciner of the Lyons kiln, the two modifications involving OE and geometry modification for mixing enhancement have the potential to reduce NOx emissions while increasing production. Historical CEMS data including NOx emissions has been collected before, during and after the two modifications discussed herein and theoretically this data could be used to assist in an evaluation of the quantitative

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[53] Letter to Ram Seetharam, April 29, 1998, CEMEX_NEIC_00008479.

[54] Expert Report of John Hofmann, Table 5-3, May 10, 2011.

[55] K. Peray, The Rotary Cement Kiln, CHS Press, 1986, pp. 81-82.

[56] Expert Report of John Hofmann, Table 5-1, May 10, 2011.

impact of each modification on NOx emissions. However, there are a large number of factors that complicate this interpretation including operator preferences, coal properties, raw mix properties, and other equipment. Due to non-linear sensitivity and complexity of NOx formation and reduction, an extensive amount of careful data mining would be required to attempt to extract meaningful correlations and it is unlikely that the necessary information would be available, particularly for the older periods.

Figure 8 presents a summary of annual average NOx emissions factor (lbs NOx/ton clinker) as a function of clinker production.

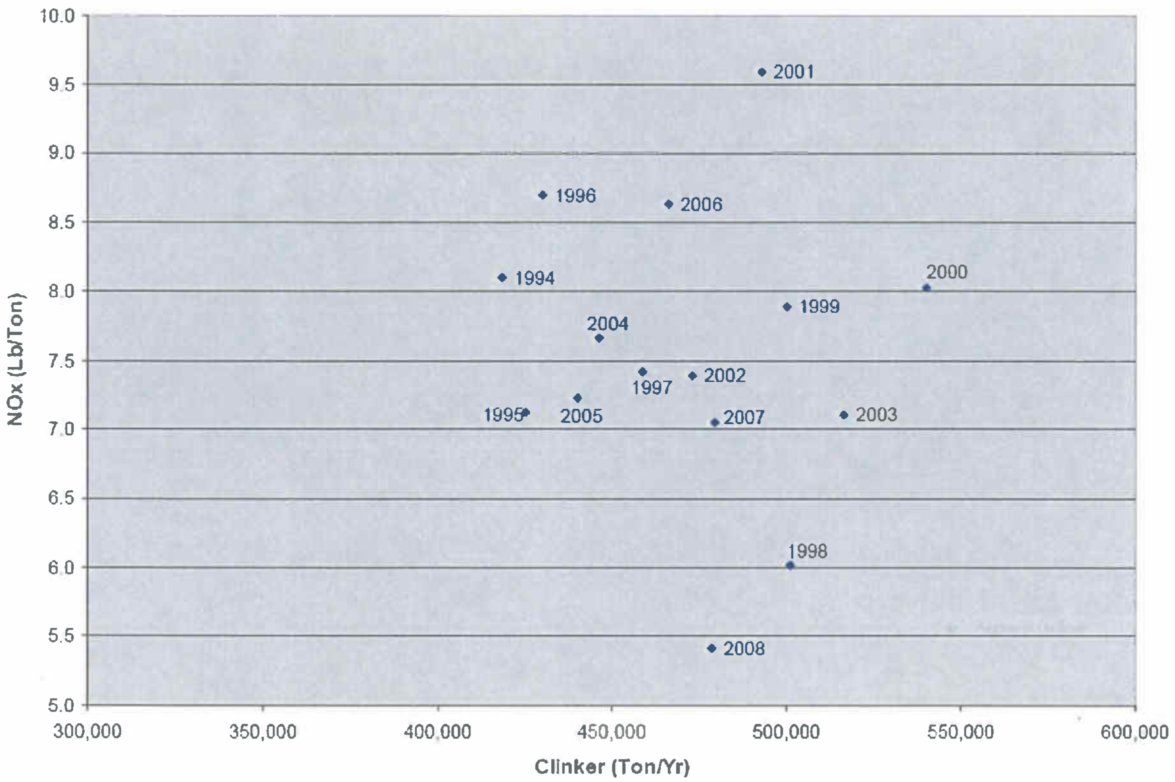

Figure 8. Annual NOx Emission Factor (lb/ton) v. Clinker Production (ton/yr).[57]

A cursory consideration of this data set indicates that there is no correlation between NOx emissions factors and clinker production. In relation to peak values, NOx emissions factors vary approximately 44% while clinker production varies approximately 22% over the same period. Therefore, it would be illogical to make any assumptions regarding the impact of individual changes in operation, fuel, raw mix, or equipment based on simple emissions factors. Based on fundamental considerations, the two modifications discussed herein could reasonably be expected to reduce NOx emissions while increasing fuel efficiency and clinker production. Carefully conducted testing or data mining under well documented periods of time between

[57] Expert Report of Phyllis Fox, p. 26, March, 2011.

instances of any significant equipment modification or fuel/raw mix changes would provide greater certainty in reaching this conclusion, but a simple consideration of clinker production and application of constant emissions factor is not a reasonable approach to make conclusions regarding NOx emissions due to equipment modifications to the Lyons cement kiln.

Appendix A: List of Documents Considered in Preparation of This Report

DOCUMENTS

1997 Oxygen Injection Test Results, CMXLYO00612454

1998 Standard Diary, CMXLYO00065175-84

A. Kokkinos, J. Cichanowicz, D. Eskinazi, J. Stallings, G. Offen, "NOx Controls for Utility Boilers: Highlights of the EPRI July 1992 Workshop," J. Air and Waste Management Association, 42, pp. 1498 – 1505, 1992

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Burning and Cooling Flowchart, June 28, 2002, CEMEX_NEIC_00005458-59

Calciner Diagram, CMXLYO00402670

CEMEX Lyons Facility, Estimated Clinker Production - Historical Summary, CMXLYO00628849-852

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Letter from Dave Repasz to Bruce Tompkins, Re: Lyons Visit, November 10, 1997, CEMEX_NEIC_00003282-84

Letter from John Lohr to Ram Seetharam, April 29, 2008, CEMEX_NEIC_00008479-8485

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Memorandum from John Lohr to Dave Repasz and Bruce Ballinger, Re: Lyons Kiln Production and Oxygen Use, January 4, 1998, CEMEX_NEIC_00008526-28

Memorandum from John Lohr to Dave Repasz, Re: 1998 Operating Budget Revisions, November 13, 1997, CMXLYO00045758-63

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Memorandum from John Lohr to Steve Bryan, Re: Improvement Project Update, July 20, 2000, CMXLYO00045597-98

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Operations Report, April 9, 1998, CMXLYO00085751-54

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Project Outline for Oxygen Enrichment Test at Lyons, March 1997 CMXLYO00612385-405

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S. Chen, J. McCarthy, W. Clark, M. Heap, W. Seeker, D. Pershing "Bench and Pilot Scale Process Evaluation of Reburning for In-furnace NOx Reduction," 21st Symposium (International) on Combustion, pp. 1159-1169. The Combustion Institute, 1986

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DEPOSITIONS

Deposition of Kevin Kelley, December 16, 2010 (and exhibits)

Deposition of John Lohr, October 11, 2010 (and exhibits)

Deposition of Randall Wiley, November 10, 2010 (and exhibits)

Deposition of Stephen Goodrich, December 9, 2010 (and exhibits)

Deposition of Thomas Boardman, August 12, 2010 (and exhibits)

EXPERT REPORTS

Expert Report of Gerald Young, prepared on behalf of CEMEX, Inc. in United States v. CEMEX, Inc., Civil Action No. 09-cv-0019-MSK-MEH, U.S. District Court for the District of Colorado, March 10, 2011 (and documents cited therein)

Expert Report of John Hofmann, prepared on behalf of CEMEX, Inc. in United States v. CEMEX, Inc., Civil Action No. 09-cv-0019-MSK-MEH, U.S. District Court for the District of Colorado, March 10, 2011 (and documents cited therein)

Expert Report of Phyllis Fox, Ph.D., prepared on behalf of the United States in United States v. CEMEX, Inc., Civil Action No. 09-cv-0019-MSK-MEH, U.S. District Court for the District of Colorado, March 1, 2011 (and documents cited therein)

Expert Report of Dr. Ranajit (Ron) Sahu, prepared on behalf of the United States in United States v. CEMEX, Inc., Civil Action No. 09-cv-0019-MSK-MEH, U.S. District Court for the District of Colorado, March 1, 2011 (and documents cited therein)

Expert Report of Walter L. Greer, prepared on behalf of the United States in United States v. CEMEX, Inc., Civil Action No. 09-cv-0019-MSK-MEH, U.S. District Court for the District of Colorado, March 1, 2011 (and documents cited there in)

Appendix B: Author Qualifications

Dr. Davis has worked in the fields of combustion and material science and technology for 24 years and has a broad range of experience in industrial, academic, and national lab environments. At Reaction Engineering International he has led projects in combustion research and product and technology development. His efforts have included successful programs spanning a range of technical areas including: NOx/fuel utilization/corrosion relationships in coal- and biomass-fired boilers, Computational modeling, mineral ore conversion in rotary kilns and flash smelters, high temperature corrosion monitoring and modeling, heterogeneous catalysis/adsorption/reaction, and fluidized bed applications. While working for Sandia National Laboratories, Dr. Davis served as principal investigator of research programs in the areas of droplet and char combustion including work acknowledged by the Combustion Institute with their Silver Medal. His Master's and Ph.D. research, completed at Princeton University under Professor Irvin Glassman, developed a thermodynamic foundation and experimental evidence establishing the feasibility of a flame synthesis technique for producing nanoscale particles of several important classes of ceramics. Dr. Davis has authored several publications on these topics.

Education

Ph.D.	Princeton University	Mechanical & Aerospace Eng.	1992
M.A.	Princeton University	Mechanical & Aerospace Eng.	1989
B.S.	University of Texas at Austin	Mechanical Engineering	1987

Experience

Reaction Engineering International

- Partner and Manager, Business Development (2002 - Present)
- Manager, Engineering Research & Development (1998 - 2002)
- Senior Engineer, (1995 - 1998)

    - ➢ Project lead and technical roles on governmental and industrial programs sponsored by DOE, NASA, EPA, DOD and a variety of industrial concerns. Technical areas of focus include the following: Relationships between NOx and unburned carbon during combustion of pulverized fuel with emphases on furnace operating parameters, burner design and reburning. Computational simulation and code development have focused on wall-fired, tangential and grate fired boilers as well as test furnaces.
    - ➢ Development and validation of analytical and empirical techniques for computational modeling of rotary kilns for mineral processing.
    - ➢ Development of corrosion modeling and monitoring tools for application in coal-fired boilers.
    - ➢ Development, design, and testing of a fluidized-bed incinerator for use with high alkali content waste in a closed-loop environment for long term missions.

- Established the feasibility of a fluorite-type metal oxide catalyst for reduction of SOx, NOx, and VOCs.
- Development of single particle models for the reaction of copper and lead concentrates in flash smelting.
- Directed experimental investigation establishing the kinetics and feasibility of mercury adsorption using unburned carbon in flyash.

University of Utah

- Assistant Research Professor (1995 - 2002)
  - Oversight for various university programs involving graduate student thesis work in the areas of coal combustion, air toxics, and incineration.

Sandia National Labs, Center for Combustion/Materials Science & Technology

- Technical Staff (1992 - 1995)
  - Experimental study involved fundamental research evaluating the potential on biomass-derived pyrolysis oils for as a replacement for fossil fuels. Advanced imaging techniques and a state of the art droplet generation system were developed to study reaction rates, microexplosion phenomena, and coke formation. Assisted and eventually directed investigation of coal combustion to complete carbon conversion.
  - Advanced characterization techniques for the purpose of explaining reactivity loss of unburned carbon in partially reacted char particles. Developed HRTEM image analysis technique for evaluation of char crystallization. Developed and applied an FTIR spectroscopy system capable of measuring nitrogen species concentrations below ppm levels.
  - Developed a program to evaluate co-firing of propellants and explosives with traditional fuels as an acceptable alternative to open burn/detonation or incineration (NOx/SOx emissions, flame stability, ash characterization) in Sandia's pilot-scale facility designed for simulation of a range of practical combustors.

Selected Honors and Awards

- Reviewer for Combustion Institute, Combustion & Flame, Energy & Fuels, Progress in Energy and Combustion Science, Combustion Science & Technology, Thermal Science and Engineering Applications, AIChE, ASME, Clean Air, Chemical Engineering Science
- National Research Council Committee for review of DOE's Industrial Technology Program, 2004-2005
- Emerald Literati Highly Commended Paper Award in Anti-Corrosion Methods and Materials, 2004
- American Chemical Society Subcommittee on Fuel Combustion Chemistry, 1997 - present
- NASA LBJ Group Achievement Award for outstanding accomplishments, 1998
- Silver Medal of the Combustion Institute, 1996
- Air Force Research in Aeropropulsion Technology Graduate Fellowship, 1991
- University of Texas National Merit Scholar, 1983-1987

Appendix C: Author Publications, Papers and Invited Lectures In the Past Ten Years

A. Fry, B. Adams, K. Davis, D. Swensen, S. Munson, W. Cox "An investigation into the likely impact of oxy-coal retrofit on fire-side corrosion behavior in utility boilers," International Journal of Greenhouse Gas Control, submitted for publication, 2011.

K. Davis, H. Shim, M. Geier, C. Shaddix, "On the use of single-film models to describe the oxy-fuel combustion of pulverized coal char," 36th International Technical Conference on Coal Utilization & Fuel Systems, Clearwater, FL, June 2011.

A. Fry, B. Adams, K. Davis, D. Swensen, S. Munson, W. Cox "Fire-side corrosion of heat transfer surface materials for air- and oxy-coal combustion," 36th International Technical Conference on Coal Utilization & Fuel Systems, Clearwater, FL, June 2011.

K. Davis "A CFD Analysis of Stove Oxygen Enrichment in a Hot Blast Stove," Energy and Utilities Committee of the American Institute of Steel Technology, Purdue-Calumet, October, 26, 2010.

K. Davis "Biomass Power Generation: Co-firing & Dedicated Firing Challenges," Air Pollution Control Round Table, Reinhold Environmental, Charlotte, NC, July 19-20, 2010.

K. Davis, S. Lokare and H. Shim, R. Minkara, S. Harding "A Physical Coal Upgrading Process and Its Impact on NOx Emissions," 33rd International Technical Conference on Clean Coal & Fuel Systems, Clearwater, FL, June 2010.

H. Wang, K. Davis, C. Senior, "CFD Modeling of Rotary Kilns for Pyro-processing Applications," International Conference on Thermal Treatment Technologies - IT3, May 17-20, 2010.

I. Jameel, K. Davis, "A New Air System for Improving Straw Combustion in a Grate Fired Boiler," International Biomass Conference, Minneapolis, MN, May 4-6, 2010.

K. Davis "Sustainable Coal Combustion," Invited Lecture at the Korea-US International Workshop on Green Science & Technology, Daejon, South Korea, December 3, 2009.

K. Davis "Coal, Biomass, Co-firing, Alternative Fuels: Which Way to Go?" Invited Panel Discussion, Air Pollution Control Roundtable, Reinhold Environmental, The Woodlands, TX, July 2009.

K. Davis, H. Shim, S. Harding, R. Minkara "A physical coal upgrading process and its impact on NOx emissions," 32nd International Technical Conference on Clean Coal & Fuel Systems, Clearwater, FL, May/June 2009.

K. Davis "A Real-Time Monitoring Technique for Fire-side Corrosion Characterization in Boilers" 23rd Annual ACERC Technical Conference, Provo, UT, February 2009

K. Davis "Emissions considerations affecting practical implementations of biomass combustion systems for power/steam generation." Colorado School of Mines ChE Fall Seminar Series, September, 2008.

A. Fry, K. Davis, M. Cremer "Advanced Layered Technology Approach (ALTA): CFD Analysis of a 180MW pulverized coal fired boiler implementation," 33rd International Technical Conference on Coal Utilization & Fuel Systems, Clearwater, FL, June 2008.

K. Davis "Firing Biomass – Plant Considerations, International Conference on Clean and Secure Energy, American Flame Research Committee, Park City, UT, May 2008.

K. Davis "Workshop on Firing Biomass – Plant Considerations" NOx Roundtable and Expo, Reinhold Environmental, Richmond, VA, February 2008.

A. Fry, K. Davis. D. Wang, T. Gale "CFD Modeling of an oxy-coal combustion retrofit," Electric Utilities Environmental Conference, Tucson, AZ, January, 2008.

A. Fry, K. Davis, M. Cremer "CFD Modeling of an oxy-coal combustion retrofit," Electric Utilities Environmental Conference, Tucson, AZ, January, 2008.

J. R. Valentine, H. Shim, K.A. Davis, S. Seo, T. Kim "CFD evaluation of waterwall wastage in coal-fired utility boilers," Energy & Fuels 2007, 21, 242-249

H. Shim, J. Valentine, and K. Davis, S. Seo, E. Kim, T. Kim "Investigation of fire-side corrosion in a pilot-scale test furnace," 32nd International Technical Conference on Coal Utilization & Fuel Systems, Clearwater, FL, June 2007.

A. Fry, K. Davis, M. Cremer "NOx Control Developments with ALTA in Pulverized Coal Units," 32nd International Technical Conference on Coal Utilization & Fuel Systems, Clearwater, FL, June 2007.

K. Davis "Application of NOx Control Technologies," Applied Combustion Technology Short Course, Salt Lake City, UT, May 2007.

H. Shim, J. Valentine, K. Davis, S. Seo, E. Kim, T. Kim "Fire-side corrosion study in a coal-fired pilot-scale test furnace," Impacts of Fuel Quality on Power Production, Snowbird, UT, October/November 2006.

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K. Davis, D. Wang, M. Cremer, E. Eddings "Advanced Approach to Layering of In-furnace NOx Control Technologies," 31st International Technical Conference on Coal Utilization & Fuel Systems, Clearwater, FL, May 2006.

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K. Davis "Advanced modeling techniques for advanced power generation and industrial combustion," Korea Advanced Institute of Science and Technology, Combustion Engineering Research Center Seminar, Daejeon, South Korea, June, 2005.

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